

CRICKET WORLD CUP 2027

Introduction

You have been assigned the task of designing the ICC Cricket World Cup 2027 network infrastructure. The tournament will be hosted in South Africa. For simplicity, you have been asked to design the team hotel networks for five participating countries: Bangladesh, India, Pakistan, Sri Lanka, and Afghanistan. Each team will be staying in a dedicated hotel, and each hotel must be connected through a reliable network to ensure smooth communication during the tournament. The total number of staff/media personnel required for each hotel is given below:

- Bangladesh → 180 Hosts
- India → 300 Hosts
- Pakistan → 220 Hosts
- Sri Lanka → 150 Hosts
- Afghanistan → 120 Hosts

Requirements

Topology Design & Server Setup

- Each country's hotel is represented by a router.
- Afghanistan's hotel is only connected to Pakistan's hotel.
- Pakistan, Sri Lanka, and Bangladesh's hotels are connected to one another via a switch (shared network segment).
- There is a loop among the Asian teams' hotels:
 - Bangladesh is directly connected to India
 - India is connected to Sri Lanka
 - Sri Lanka is connected to Pakistan
 - Pakistan is connected back to Bangladesh
- All hotels have their own Email Servers.
 - Hostnames should follow this format:
 - mail.bangladesh.com
 - mail.india.com
 - mail.pakistan.com, etc.
- Only India (www.india.cricket.com) and Bangladesh (www.bangladesh.cricket.com) have dedicated Web Servers.
 - Configure them so all devices can access these websites.
- Only Pakistan's hotel has the central DNS server.
 - Configure DNS so all devices across the network can resolve domain names.
- Bangladesh's router acts as the central DHCP server for all hotels except:
 - India and Afghanistan → have their own dedicated DHCP servers.
- Each hotel contains:
 - One printer (static IP)
 - Two PCs (representing all hosts, using DHCP)

Addressing

- Select the student ID of the first group member.
- Take the last 4 digits, split into two parts (2 digits each).

- These form the first two octets of the network.

Example:

ID = 20231234 → Last 4 digits = 1234

→ Network = 12.34.0.0/16

- Use VLSM (Variable Length Subnet Masking) to divide the network efficiently based on host requirements.
- Each hotel must have its own subnet.
- Rules:
 - PCs → Dynamic IP (DHCP)
 - Routers, Servers, Printers → Static IP

Routing

- Bangladesh, India, and Sri Lanka use RIP protocol to share routing information dynamically.
- Bangladesh's router is aware of all networks (acts as central intelligence).
- Afghanistan can only reach other networks through Pakistan.
 - Configure a default route.
- Pakistan and India use static routing to communicate with other networks.
- If the path between India and Bangladesh fails, India must send packets via an alternate route using recursive routing.

Additional Assumptions

- All links between routers are point-to-point unless connected via a switch.
- Appropriate IP ranges must be reserved for servers and infrastructure devices.
- Network should ensure full connectivity (end-to-end ping success).

Deliverables

You must submit the following:

1. Work Distribution

Clearly mention who completed:

- Topology
- VLSM
- Configuration
- Testing

2. Packet Tracer File

- Complete .pkt / .pka file
- All configurations must be working

3. Network Topology Diagram

- Clean, labeled diagram
- Show:
 - Routers
 - Links
 - Network addresses (using notes)

4. A PDF Document Including:

- VLSM Tree
- IP Address Table
- Router Configuration Commands